

P7 – AI-Assisted Crop Type Mapping using Sentinel-2 NDVI and Random Forest Classification

Project Report Submitted in fulfillment of
the Requirements for the Award of the Internship of Summer Training Program

Subject Name

Remote Sensing And Gis In Python

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P7-AI-Assisted Crop Type Mapping using Sentinel-2 NDVI and Random Forest Classification

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Date of Submission:

2nd July 2026

Subject:

Remote sensing and Gis in python

Abstract:

This project focuses on AI-assisted crop type mapping using Sentinel-2 satellite imagery and Random Forest machine learning classification. NDVI was generated using Red and Near Infrared bands to analyze vegetation health and distribution. The processed raster data was classified into vegetation density categories using machine learning techniques. The project demonstrates the integration of GIS, remote sensing, and artificial intelligence for agricultural monitoring and crop analysis.

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1. Introduction

Remote sensing and Geographic Information Systems (GIS) play a major role in modern agricultural monitoring and environmental analysis.

Satellite imagery provides valuable information about vegetation, land cover, soil conditions, and crop health over large geographical areas. With the advancement of Artificial Intelligence (AI) and Machine Learning (ML), remote sensing applications have become more efficient and automated.

This project focuses on AI-assisted crop type mapping using Sentinel-2 satellite imagery and Random Forest machine learning classification.

Sentinel-2 imagery provides high-resolution multispectral data suitable for vegetation analysis and agricultural monitoring. In this project, the Normalized Difference Vegetation Index (NDVI) was generated using the Red and Near Infrared (NIR) bands of Sentinel-2 imagery to analyze vegetation distribution and crop density.

The generated NDVI data was cleaned and processed using Python libraries such as Rasterio, NumPy, and Matplotlib. Machine learning techniques were then applied to classify vegetation regions into different crop density categories. A Random Forest classifier was used to automate the classification process and generate an AI-assisted crop classification map.

The integration of GIS, remote sensing, and machine learning provides an effective and scalable approach for crop monitoring and agricultural analysis. The project demonstrates how AI-assisted geospatial workflows can support precision agriculture, vegetation analysis, and land cover mapping using satellite imagery and Python-based data processing techniques.

2.Objectives:

- To download and process Sentinel-2 satellite imagery
 - To generate NDVI for vegetation analysis
 - To clean and preprocess raster data
 - To apply Random Forest machine learning classification
 - To generate crop classification maps
 - To visualize vegetation distribution
-

3. Study Area

The study area selected for this project is the Warangal region in Telangana, India. The area consists mainly of agricultural land, vegetation cover, water bodies, and semi-urban regions, making it suitable for crop and vegetation analysis.

Sentinel-2 satellite imagery covering the study area was downloaded and processed for NDVI generation and AI-assisted crop classification. The region was selected due to its agricultural importance and the availability of suitable multispectral satellite data for remote sensing analysis.

The study area was further used for raster preprocessing, vegetation analysis, and Random Forest-based classification during the project workflow.

4. Data Used

The project uses Sentinel-2 multispectral satellite imagery acquired from the Copernicus Open Access platform. The satellite data contains multiple spectral bands useful for vegetation and land cover analysis.

The data used for this calculation is from 2023 jan to 2023 dec. Cloud cover is <10%.

The following Sentinel-2 bands were used in this project:

- B04 (Red Band) – Used for vegetation analysis
- B08 (Near Infrared Band) – Used for NDVI generation

Additional datasets used include:

- Telangana study area boundary shapefile
- NDVI raster generated using Python
- Classified raster outputs generated using Random Forest classification



The data was processed using QGIS, Python, Rasterio, NumPy, Matplotlib, and Scikit-learn libraries.

5. Methodology:

1. Download Sentinel-2 imagery from Copernicus
2. Extract B04 (Red) and B08 (NIR) bands

Python code for loadings bands:

```
# File paths

b04_path = r"C:\Users\visha\OneDrive\Desktop\Crop-Type-Mapping\data\raw\S2B_MSIL2A_20231025T050859_N0510_R019_T44QMF_20241102T215233.SAFE\GRANULE\L2A_T44QMF_A034651
b08_path = r"C:\Users\visha\OneDrive\Desktop\Crop-Type-Mapping\data\raw\S2B_MSIL2A_20231025T050859_N0510_R019_T44QMF_20241102T215233.SAFE\GRANULE\L2A_T44QMF_A034651

shapefile_path = r"C:\Users\visha\OneDrive\Desktop\Crop-Type-Mapping\data\processed\telangana_32644.shp.gpkg"
```

Python

```
gdf = gpd.read_file(shapefile_path)

print(gdf.head())
```

Python

```

  GID_1 GID_0 COUNTRY      NAME_1 VARNAME_1 NL_NAME_1 TYPE_1 \
0  IND_2_1  IND   India  Andhra Pradesh      NA      NA  State
1  IND_32_1  IND   India    Telangana      NA      NA  State

  ENGTYPE_1 CC_1 HASC_1  ISO_1 \
0    State   NA  IN.AP  IN-AP
1    State   NA  IN.TG  IN-TG

      geometry
0  MULTIPOLYGON (((254857.386 1443258.407, 254065...
1  MULTIPOLYGON (((308413.906 1795289.02, 306746....
```

```
# Read Red band

with rasterio.open(b04_path) as red_src:
    red = red_src.read(1).astype('float32')
    profile = red_src.profile

# Read NIR band

with rasterio.open(b08_path) as nir_src:
    nir = nir_src.read(1).astype('float32')
```

Python

3. Generate NDVI

4. Clean invalid NDVI values

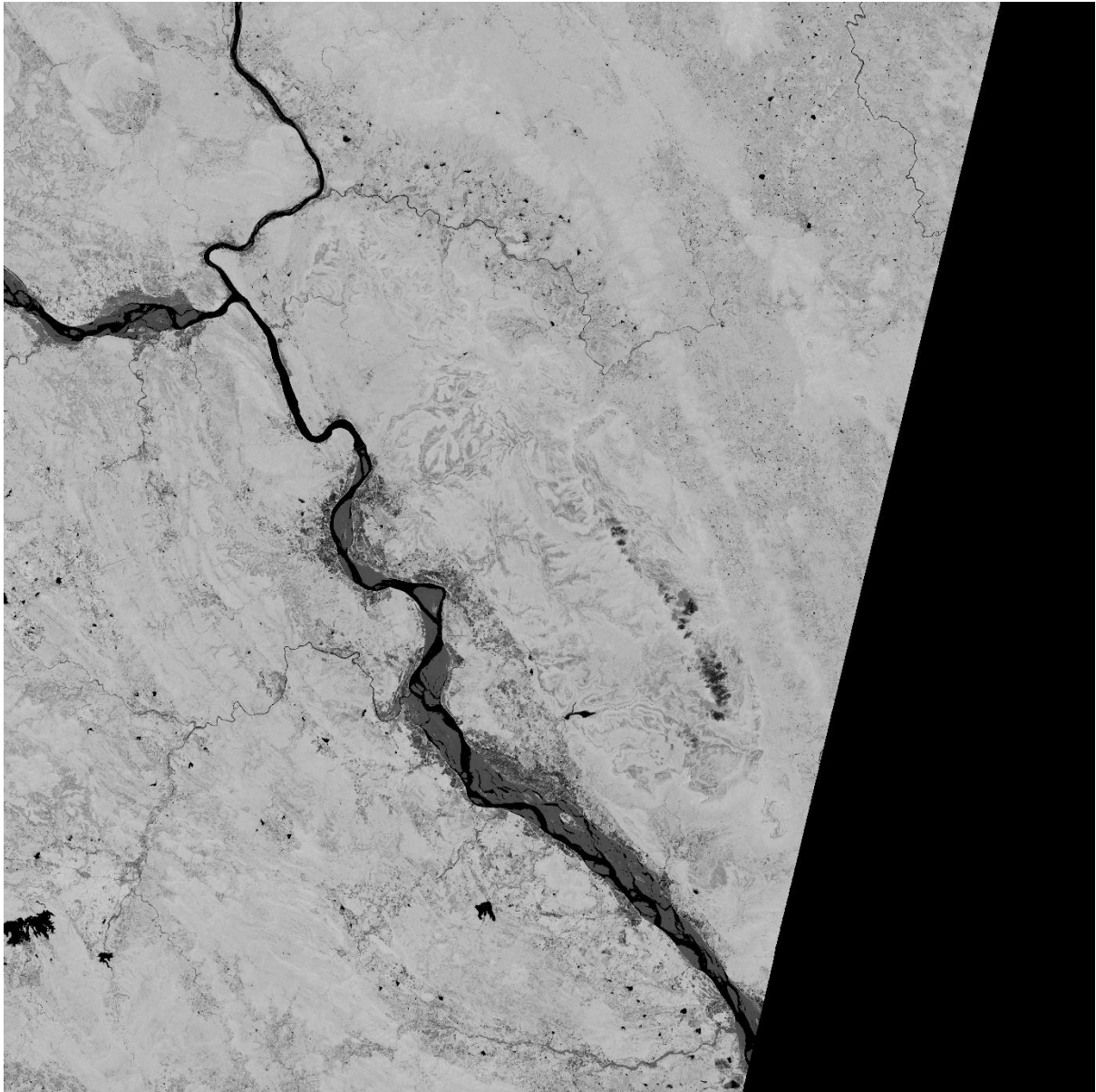
```
# NDVI calculation

ndvi = (nir - red) / (nir + red + 1e-10)

plt.figure(figsize=(10,10))
plt.imshow(ndvi, cmap='RdYlGn')
plt.colorbar(label='NDVI')
plt.title("NDVI Warangal 2023")
plt.show()
```

Code for ndvi generation.

5. Prepare raster data for machine learning



6. Train Random Forest classifier
 7. Generate classified crop map
 8. Visualize classification outputs
-

6. NDVI Calculation

The Normalized Difference Vegetation Index (NDVI) is a widely used vegetation index in remote sensing for analyzing vegetation health and density. NDVI is calculated using the Red and Near Infrared (NIR) spectral bands obtained from Sentinel-2 satellite imagery.

The following Sentinel-2 bands were used:

- B04 – Red Band
- B08 – Near Infrared Band

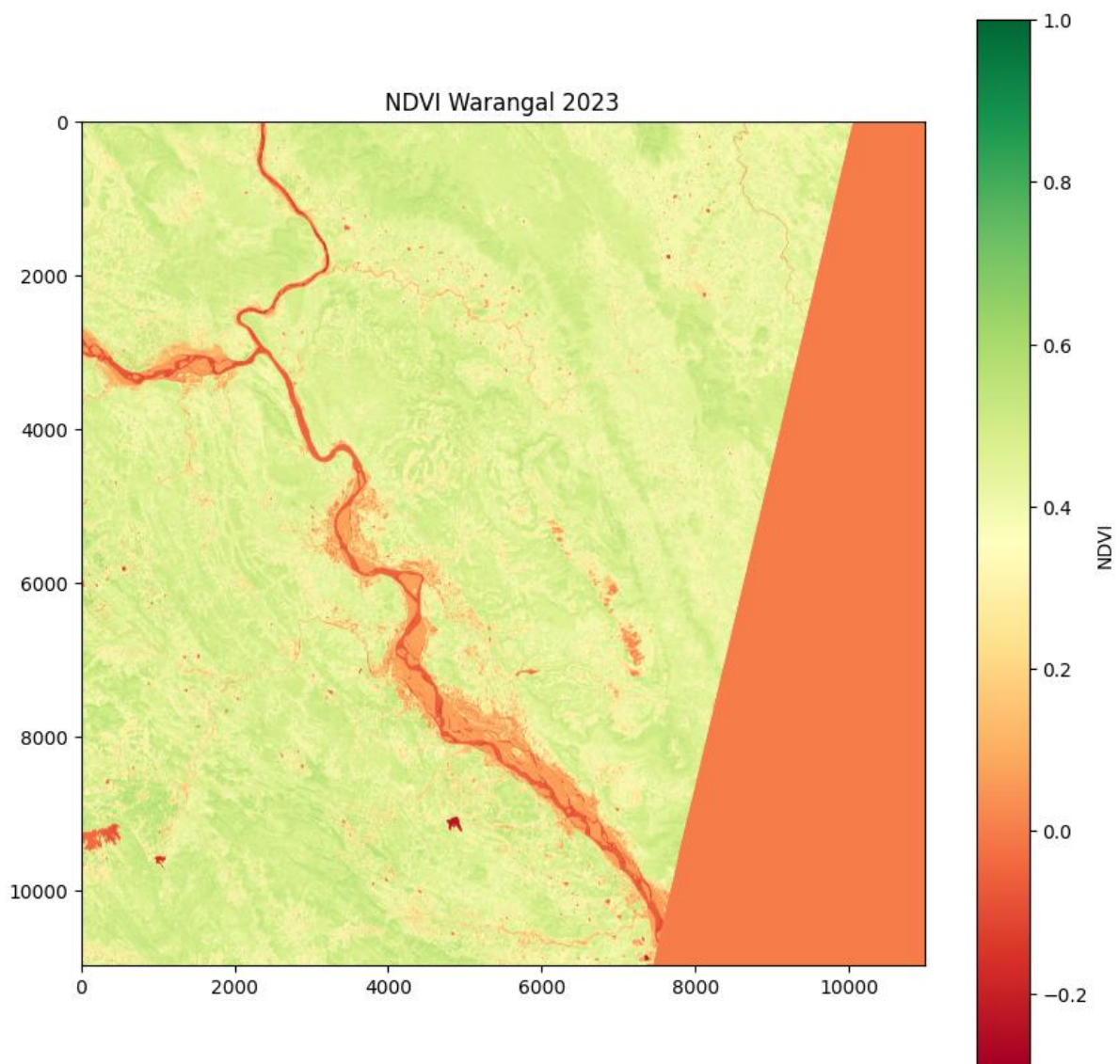
The NDVI formula used in this project is:

$$\text{NDVI} = \frac{\text{NIR} - \text{Red}}{\text{NIR} + \text{Red}}$$

NDVI values range from -1 to +1. Higher NDVI values represent healthy and dense vegetation, while lower values represent barren land, water bodies, or non-vegetated regions.

The NDVI raster was generated and processed using Python libraries such as Rasterio, NumPy, and Matplotlib for further machine learning analysis.

Output:



7. AI-Assisted Random Forest Classification

Random Forest is a supervised machine learning algorithm commonly used for classification and prediction tasks. In this project, Random Forest classification was applied to perform AI-assisted crop and vegetation mapping using NDVI raster data.

The cleaned NDVI raster was converted into machine learning input features, and vegetation categories were created based on NDVI ranges. The Random Forest model was then trained using the prepared dataset to classify vegetation density into different classes such as non-vegetation, moderate vegetation, and dense vegetation.

The model was implemented using the Scikit-learn library in Python. After training, the classifier generated a classified crop map representing vegetation distribution across the study area.

The use of Random Forest improved the automation and efficiency of crop classification and demonstrated the integration of Artificial Intelligence with remote sensing and GIS workflows.

```
import rasterio
import numpy as np
import matplotlib.pyplot as plt

from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import train_test_split
from sklearn.metrics import classification_report
from sklearn.metrics import confusion_matrix
print("Libraries loaded!")
```

Libraries loaded!

Libraries used for ml work.

```
# -----
# CELL 3 - CLEAN NDVI
# -----

ndvi_clean = np.where(ndvi > 1, np.nan, ndvi)
ndvi_clean = np.where(ndvi_clean < -1, np.nan, ndvi_clean)

print("NDVI cleaned!")

plt.figure(figsize=(10,10))
plt.imshow(ndvi_clean, cmap='RdYlGn', vmin=-1, vmax=1)
plt.colorbar(label='NDVI')
plt.title("Clean NDVI Map")
plt.show()
```

ML code flow ,trained 50000 samples.

```
# =====  
# CELL 5 - PREPARE DATA FOR ML  
# =====  
  
X = ndvi_clean.reshape(-1,1)  
  
valid_mask = ~np.isnan(X[:,0])  
  
X_valid = X[valid_mask]  
  
# Random sample  
  
sample_size = 50000  
  
indices = np.random.choice(  
    X_valid.shape[0],  
    sample_size,  
    replace=False  
)  
  
X_valid = X_valid[indices]  
  
print(X_valid.shape)
```

(50000, 1)

```
# =====  
# CELL 6 - CREATE LABELS  
# =====  
  
# 0 = Non vegetation  
# 1 = Moderate vegetation  
# 2 = Dense vegetation  
  
y = np.where(X_valid[:,0] > 0.5, 2, 0)  
y = np.where(X_valid[:,0] > 0.2, 1, y)  
print(np.unique(y))
```

[0 1]

```
# =====  
# CELL 7 - TRAIN TEST SPLIT  
# =====  
  
X_train, X_test, y_train, y_test = train_test_split(  
    X_valid,  
    y,  
    test_size=0.3,  
    random_state=42  
)  
  
print(X_train.shape)  
print(X_test.shape)
```

```
# =====  
# CELL 8 - RANDOM FOREST MODEL  
# =====  
  
rf_model = RandomForestClassifier(  
    n_estimators=100,  
    random_state=42  
  
)  
  
rf_model.fit(X_train, y_train)  
  
print("Random Forest trained!")
```

Random Forest trained!

Code:

```
# =====  
# CELL 9 - PREDICTIONS  
# =====  
  
y_pred = rf_model.predict(X_test)  
  
print("Prediction completed!")
```

Prediction completed!

Code:

```
# =====
# CELL 10 - ACCURACY REPORT
# =====

print("Classification Report:\n")

print(classification_report(y_test, y_pred))

print("\nConfusion Matrix:\n")

print(confusion_matrix(y_test, y_pred))
```

Classification Report:

	precision	recall	f1-score	support
0	1.00	1.00	1.00	1113
1	1.00	1.00	1.00	13887
accuracy			1.00	15000
macro avg	1.00	1.00	1.00	15000
weighted avg	1.00	1.00	1.00	15000

Confusion Matrix:

```
[[ 1113    0]
 [    0 13887]]
```

Code for image classification:

```
# =====
# CELL 11 - FULL IMAGE CLASSIFICATION
# =====

# Predict ALL valid pixels
all_predictions = rf_model.predict(X[valid_mask])

# Create empty raster
classified = np.full(X.shape[0], -1)

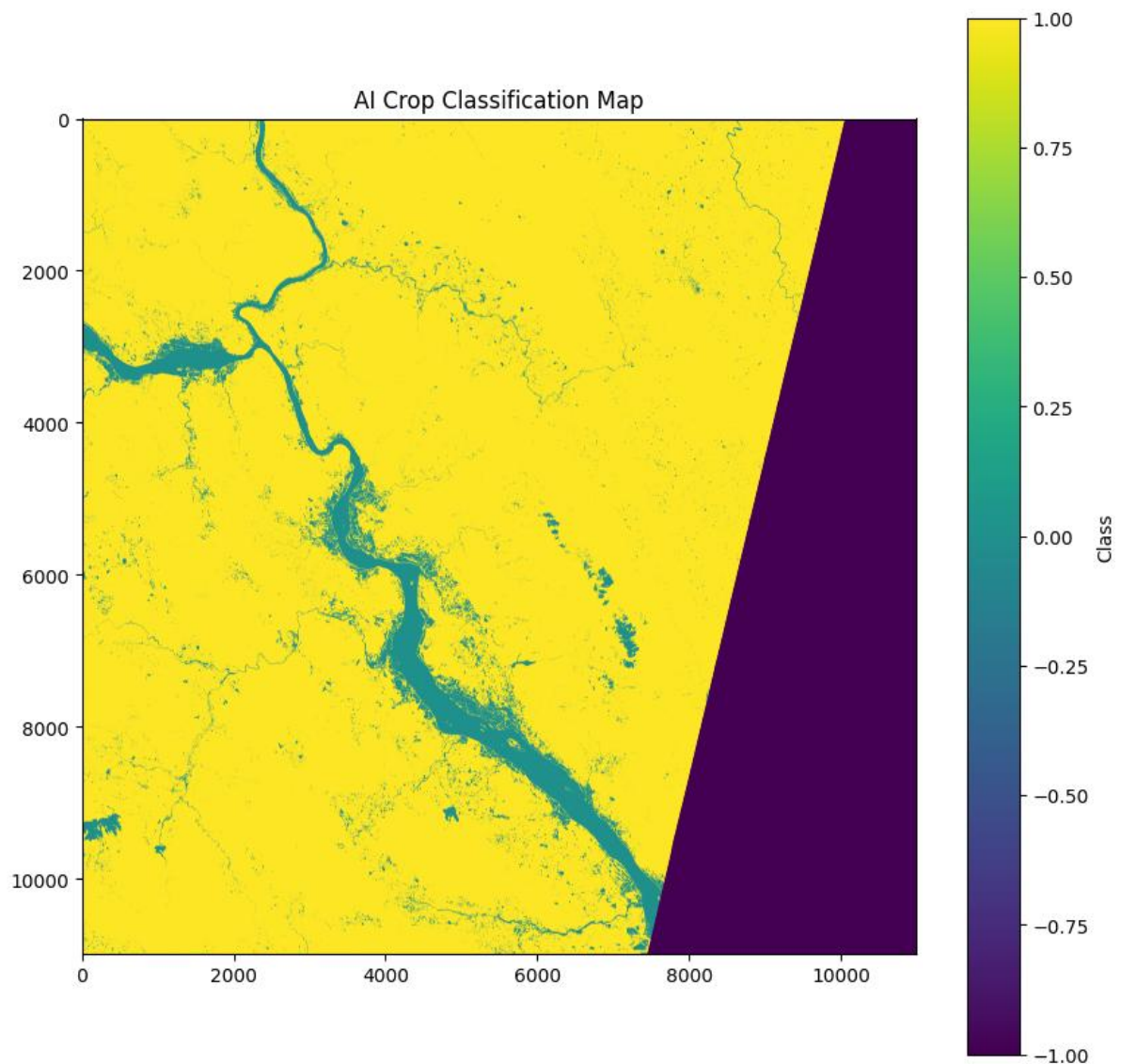
# Fill predictions
classified[valid_mask] = all_predictions

# Reshape back to image
classified_image = classified.reshape(ndvi_clean.shape)

print("Full classification done!")
```

Full classification done!

Ai classification map:



Class distribution:

```
# =====  
# CELL 14 - SHOW CLASS DISTRIBUTION  
# =====  
  
unique, counts = np.unique(classified_image, return_counts=True)  
print("Class Distribution:\n")  
  
for u, c in zip(unique, counts):  
    print(f"Class {u}: {c} pixels")
```

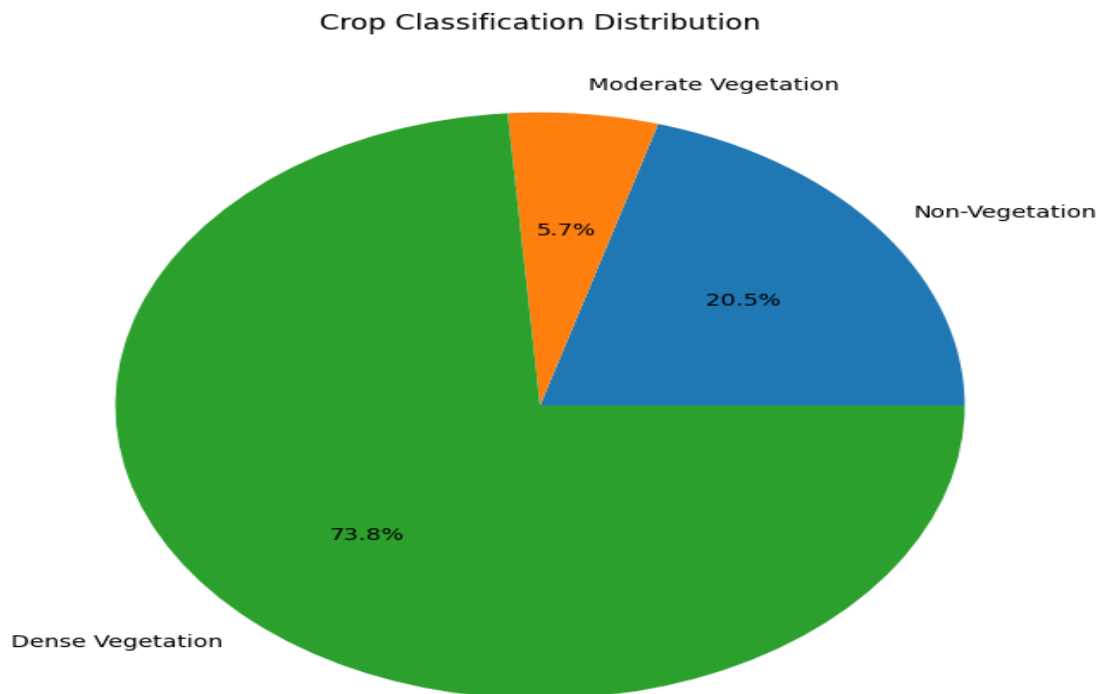
Class Distribution:

Class -1: 24730680 pixels
Class 0: 6874610 pixels
Class 1: 88955110 pixels

Pie chart code:

```
# =====  
# CELL 15 - PIE CHART  
# =====  
  
labels = [  
    'Non-Vegetation',  
    'Moderate Vegetation',  
    'Dense Vegetation'  
]  
  
pixel_counts = counts  
  
plt.figure(figsize=(8,8))  
  
plt.pie(  
    pixel_counts,  
    labels=labels,  
    autopct='%1.1f%%'  
)  
  
plt.title("Crop Classification Distribution")  
  
plt.show()
```

Output:



8. Results and Discussion

The project successfully generated NDVI maps and AI-assisted crop classification outputs using Sentinel-2 satellite imagery and Random Forest classification. The NDVI analysis helped identify vegetation density and crop distribution across the study area.

The generated NDVI maps showed higher vegetation values in agricultural and densely vegetated regions, while lower values represented barren land and non-vegetated areas. The Random Forest classifier further categorized the raster data into different vegetation classes, improving the visualization and interpretation of crop distribution patterns.

The classified outputs and graphical visualizations demonstrated the effectiveness of combining remote sensing, GIS, and machine learning techniques for agricultural analysis. The project also highlighted the usefulness of Python-based geospatial workflows for raster processing and AI-assisted land cover classification.

```
# =====  
# CELL 17 - FINAL SUMMARY  
# =====  
  
print("=====")  
print(" AI CROP MAPPING PROJECT COMPLETE ")  
print("=====")  
  
print("\nOutputs Generated:")  
  
print("1. Clean NDVI raster")  
print("2. Classified crop map")  
print("3. Classification graph")  
print("4. Random Forest model output")  
  
print("\nProject completed successfully!")
```

```
=====  
AI CROP MAPPING PROJECT COMPLETE  
=====  
  
Outputs Generated:  
1. Clean NDVI raster  
2. Classified crop map  
3. Classification graph  
4. Random Forest model output  
  
Project completed successfully!
```


9. Conclusion

The project successfully demonstrated AI-assisted crop type mapping using Sentinel-2 satellite imagery, NDVI analysis, and Random Forest machine learning classification. The integration of remote sensing, GIS, Python, and machine learning techniques helped generate vegetation and crop classification maps efficiently.

NDVI analysis provided useful information about vegetation health and distribution within the study area. The Random Forest classifier further improved the automation of vegetation classification and land cover analysis.

The project highlights the importance of AI-assisted geospatial workflows in agricultural monitoring and demonstrates how satellite imagery and machine learning can be combined for effective crop analysis and vegetation mapping.

10. Future Scope

The project can be further improved by using multi-temporal satellite imagery for seasonal crop monitoring and change detection. Additional spectral bands and vegetation indices can also be integrated to improve classification accuracy.

Advanced machine learning and deep learning techniques such as Convolutional Neural Networks (CNNs) can be applied for more detailed crop type identification and automated land cover mapping. Field survey data and ground truth information can also be incorporated for better validation and accuracy assessment.

The workflow developed in this project can be extended for large-scale agricultural monitoring, precision farming, drought assessment, and smart farming applications using AI-assisted geospatial technologies.

11. References

1. Sentinel-2 User Handbook, European Space Agency (ESA)

2. Copernicus Open Access Hub – Sentinel Satellite Data Platform
3. QGIS Documentation – Open Source Geographic Information System
4. Rasterio Documentation – Python Raster Processing Library
5. Scikit-learn Documentation – Machine Learning in Python
6. NumPy and Matplotlib Documentation – Scientific Computing and Visualization Libraries
7. Lillesand, T., Kiefer, R., & Chipman, J. (Remote Sensing and Image Interpretation)
8. Research papers and online resources related to NDVI, Remote Sensing, GIS, and Random Forest Classification